

ELTE IK - Computer Science BSc

Final Exam Topics

5. Basics of probability and statistics

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Basics of probability and statistics

Discrete and continuous random variables, law of large numbers, central limit theorem. Statistical estimation, classical statistical tests.

1 Kolmogorov probability space

The triple $(\Omega, \mathcal{A}, P)$, where:

Ω nonempty set, sample space, ω elementary event.

$\mathcal{A}$ a system of subsets of Ω , $\mathcal{A} \subset 2^\Omega$, $A \in \mathcal{A}$ events, $\mathcal{A}$ a σ -algebra.
 σ -algebra:

1. $\Omega \in \mathcal{A}$
2. $A \in \mathcal{A} \Rightarrow \overline{A} = \Omega \setminus A \in \mathcal{A}$
3. $A_1, A_2, \dots \in \mathcal{A} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$

$P : \mathcal{A} \rightarrow [0, 1]$ set function, probability, for which $P(\Omega) = 1, P(A) \geq 0, \forall A \in \mathcal{A}$, and for pairwise exclusive $(A_i \cap A_j = \emptyset, i \neq j)$ events $A_1, A_2, \dots \in \mathcal{A}$ we have $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

2 Discrete and continuous random variables

- *Random variable:* $\xi : \Omega \rightarrow \mathbb{R}$ a measurable function, that is, one for which $\{\omega : \xi(\omega) < x\} \in \mathcal{A}, \forall x \in \mathbb{R}$, where $\mathcal{A}$ is a system of subsets of the sample space (Ω) . ($\omega \in \Omega$ elementary event).
- *Distribution / distribution function of a random variable:* $F_\xi(x) = P(\xi < x), \forall x \in \mathbb{R}$.

Its properties:

1. $0 \leq F_\xi(x) \leq 1$
2. monotone increasing
3. left-continuous
4. $\lim_{x \rightarrow -\infty} F_\xi(x) = 0, \lim_{x \rightarrow \infty} F_\xi(x) = 1$

2.1 Discrete random variables

Its range is at most countably infinite, that is, it consists of elements $\{x_1 \dots x_n \dots\}$. Then its distribution: $p_k := P(\xi = x_k)$.

Name	Interpretation	Distribution	EX	D^2X
indicator $Ind(p)$	Whether an event of probability p occurs or not.	$P(X = 1) = p$ $P(X = 0) = 1 - p$	p	$p(1 - p)$
geometric (Pascal) $Geo(p)$	On which trial an event of probability p first occurs.	$P(X = k) = p(1 - p)^{k-1}$ $k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
hypergeometric $Hipgeo(N, M, n)$	Sampling without replacement.	$P(X = k) = \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}$ $k = 0, 1, \dots, n$	$n \frac{M}{N}$	$n \frac{M}{N} (1 - \frac{M}{N}) (1 - \frac{n-1}{N-1})$
binomial $Bin(n, p)$	Sampling with replacement.	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$ $k = 0, 1, \dots, n$	np	$np(1 - p)$
negative binomial $Negbin(n, p)$	On which trial an event of probability p occurs for the n -th time.	$P(X = k) = \binom{k-1}{n-1} p^n (1 - p)^{k-n}$ $k = n, n + 1, \dots$	$\frac{n}{p}$	$\frac{n(1-p)}{p^2}$
Poisson $Poi(\lambda)$	Rare event.	$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$	λ	λ

2.2 Continuous random variables

A random variable ξ is absolutely continuous if there exists a function $f(x)$ for which $F(x) = \int_{-\infty}^x f(t)dt$. In this case $f(x)$ is a *density function*. ($F(x)$ is the distribution function.)

Another formulation: for $\forall a < b$ we have $P(a < \xi < b) = \int_a^b f(t)dt$, $F_\xi(x) = P(\xi < x) = \lim_{a \rightarrow -\infty} P(a < \xi < b) = \int_{-\infty}^x f(t)dt$.

Properties of the density function:

1. $f(x) = F'(x)$
2. $f(x) \geq 0$
3. $\int_{-\infty}^{\infty} f(x)dx = 1$

Name	Distribution function	Density function	EX	D^2X
uniform $E(a, b)$	$\begin{cases} 0 & x \leq a \\ \frac{x-a}{b-a} & a < x \leq b \\ 1 & b < x \end{cases}$	$\begin{cases} \frac{1}{b-a} & a < x \leq b \\ 0 & otherwise \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
exponential $Exp(\lambda)$	$\begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & otherwise \end{cases}$	$\begin{cases} \lambda \cdot e^{-\lambda x} & x \geq 0 \\ 0 & otherwise \end{cases}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
normal $N(m, \sigma^2)$	...	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-m)^2}{2\sigma^2}} \quad x \in \mathbb{R}$	m	σ^2
standard normal $N(0, 1^2)$	$\Phi(x) = \dots$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad x \in \mathbb{R}$	0	1
gamma $\Gamma(\alpha, \lambda)$	...	$\begin{cases} \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} & x \geq 0 \\ 0 & otherwise \end{cases}$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$

2.3 Concepts

- *Convolution*: X, Y independent random variables, their convolution is the random variable $X + Y$.

- *Independence*: $P(\xi_1 < x_1, \dots, \xi_n < x_n) = \prod_{i=1}^n P(\xi_i < x_i)$ or in the discrete case: $P(\xi_1 = x_1, \dots, \xi_n = x_n) = \prod_{i=1}^n P(\xi_i = x_i)$
- *Expected value*: $(\Omega, \mathcal{A}, P)$ probability space, $X : \Omega \rightarrow \mathbb{R}$ random variable, $EX = \int_{\Omega} X dP$, if it exists. In the discrete case $EX = \sum_k x_k \cdot p_k$, if absolutely convergent. In the absolutely continuous case $EX = \int_{-\infty}^{\infty} x \cdot f(x) dx$, if absolutely continuous.
- *Variance*: $D^2X = E((X - EX)^2) = EX^2 - E^2X$
- *l-th moment*: $EX^l = \int_{\Omega} x^l dP$, if it exists.
- *Standard deviation*: $DX = \sqrt{D^2X}$
- *Covariance*: $cov(X, Y) = E((X - EX)(Y - EY)) = E(XY) - EX \cdot EY$. If $cov(X, Y) = 0$, then X and Y are uncorrelated. (Remark: if two random variables are independent, then $cov(X, Y) = 0$, that is, they are uncorrelated; also $cov(X, X) = D^2X$.)
- *Correlation*: $R(X, Y) = \frac{cov(X, Y)}{DX \cdot DY}$, measures the linear relationship of two random variables. $R > 0 \rightarrow$ positive, $R < 0 \rightarrow$ negative; $R^2 \sim 1 \rightarrow$ strong, $R^2 \sim 0.5 \rightarrow$ medium, $R^2 \sim 0 \rightarrow$ weak.

3 Law of large numbers

3.1 Weak law

$X_1, X_2, \dots$ independent, identically distributed, $EX_i = m < \infty$, $D^2X_i = \sigma^2 < \infty$.

$P\left(\frac{X_1 + \dots + X_n}{n} - m \geq \varepsilon\right) \rightarrow 0$ ($n \rightarrow \infty$) for $\forall \varepsilon > 0$ (stochastic convergence).

3.2 Strong law

$X_1, X_2, \dots$ independent, identically distributed, $EX_1 = m < \infty$, $D^2X_1 = \sigma^2 < \infty$.

$\frac{X_1 + \dots + X_n}{n} \rightarrow m$ ($n \rightarrow \infty$) with probability 1.

Remark: We prove it with the Chebyshev inequality. $(\frac{\sigma^2}{n\varepsilon^2} \rightarrow 0$ ($n \rightarrow \infty$))

3.2.1 Chebyshev inequality

EX finite.

Then $P(|X - EX| \geq \lambda) \leq \frac{D^2X}{\lambda^2}$

Remark: Proof with the Markov inequality.

3.2.2 Markov inequality

$X \geq 0, c > 0$.

Then $P(X \geq c) \leq \frac{EX}{c}$

3.3 Types of convergence

$\xi_n \rightarrow \xi$, that is, ξ is convergent.

- *stochastically*: if for $\forall \varepsilon > 0$ $P(|\xi_n - \xi| \geq \varepsilon) \rightarrow 0$ ($n \rightarrow \infty$).
- *with probability 1 (almost everywhere)*: if $P(\omega : \xi_n(\omega) \rightarrow \xi(\omega)) = 1$.
- *in L^p* : if $E(|\xi_n - \xi|^p) \rightarrow 0$ ($n \rightarrow \infty$) ($p > 0$ fixed).
- *in distribution*: if $F_{\xi_n}(x) \rightarrow F_{\xi}(x)$ ($n \rightarrow \infty$) at every continuity point of the latter.

Their relationships: convergence with probability 1 and in L^p are the strongest; from these follows stochastic convergence, and from this convergence in distribution.

4 Central limit theorem

$X_1, X_2, \dots$ independent, identically distributed, $EX_1 = m < \infty$, $D^2 X_1 = \sigma^2 < \infty$.

Then $\frac{X_1 + \dots + X_n - nm}{\sqrt{n}\sigma} \rightarrow N(0, 1)$ ($n \rightarrow \infty$) in distribution, that is, $P(\frac{X_1 + \dots + X_n - nm}{\sqrt{n}\sigma} < x) \rightarrow \Phi(x)$ ($n \rightarrow \infty$).

5 Statistical space

The triple $(\Omega, \mathcal{A}, \mathcal{P})$, if $\mathcal{P} = \{P_\vartheta\}_{\vartheta \in \Theta}$ and $(\Omega, \mathcal{A}, P_\vartheta)$ is a Kolmogorov probability space for $\forall \vartheta \in \Theta$.

5.1 Concepts

- *Sample*: $\underline{\xi} = (\xi_1, \dots, \xi_n) : \Omega \rightarrow \Xi \in \mathbb{R}^n$. (ξ_i random variable)
- *Sample space*: Ξ , the set of possible values of the sample, often $\mathbb{R}^n, \mathbb{Z}^n$.
- *Sample [realization]*: $\underline{x} = (x_1, \dots, x_n)$, a concrete observation.
- *Statistic*: $T : \Xi \rightarrow \mathbb{R}^k$.
- *Fundamental theorem of statistics*: (Glivenko–Cantelli theorem) $\xi_1, \xi_2, \dots$ independent, identically distributed with distribution function F . Then for the empirical distribution function F_n we have $\sup_{-\infty < x < \infty} |F_n(x) - F(x)| \rightarrow 0$ ($n \rightarrow \infty$) with probability 1.

6 Statistical estimation

$(\Omega, \mathcal{A}, \mathcal{P})$ statistical space, $\vartheta \in \Theta$, $P_\vartheta(\xi_1 < x_1, \dots, \xi_n < x_n) = F_\vartheta(\underline{x})$

$T(\underline{\xi})$ is an *estimator* of ϑ if $T : \mathbb{R}^n \rightarrow \Theta$.

$T(\underline{\xi})$ is an *estimator* of $h(\vartheta)$ if $T : \mathbb{R}^n \rightarrow h(\Theta)$.

Unbiasedness: $T(\underline{\xi})$ is an unbiased estimator of $h(\vartheta)$ if $E_\vartheta T(\underline{\xi}) = h(\vartheta)$ for $\forall \vartheta \in \Theta$.

Asymptotically unbiased: $T(\underline{\xi})$ is asymptotically unbiased for $h(\vartheta)$ if $E_\vartheta T(\underline{\xi}) \rightarrow h(\vartheta)$ ($n \rightarrow \infty$) for $\forall \vartheta \in \Theta$.

The unbiased estimator T_1 is *more efficient* than the unbiased estimator T_2 if $D_\vartheta^2 T_1 \leq D_\vartheta^2 T_2$ for $\forall \vartheta \in \Theta$.

It is efficient if it is more efficient than every other unbiased estimator. If an efficient estimator exists, then it is unique.

- *Maximum likelihood estimation*: Likelihood function: $L(\vartheta, \underline{x}) = \begin{cases} P_\vartheta(\underline{\xi} = \underline{x}) & \text{discrete} \\ f_{\vartheta, \underline{\xi}}(\underline{x}) & \text{abs.cont.} \end{cases}$

In the independent case: $L(\vartheta, \underline{x}) = \begin{cases} \prod_{i=1}^n P_\vartheta(\xi_i = x_i) & \text{discrete} \\ \prod_{i=1}^n f_{\vartheta, \xi_i}(x_i) & \text{abs.cont.} \end{cases}$

$\hat{\vartheta}$ is the maximum likelihood estimator of the unknown parameter ϑ if $L(\hat{\vartheta}, \underline{\xi}) = \max_{\vartheta \in \Theta} L(\vartheta, \underline{\xi})$.

- *Method of moments estimation*: $\vartheta = (\vartheta_1, \dots, \vartheta_k)$, $\xi_1, \dots, \xi_n$, l -th moment: $M_l(\vartheta) = E_{\vartheta} \xi_i^l$, empirical l -th moment: $\hat{M}_l = \frac{\sum_{i=1}^n \xi_i^l}{n}$.
 $\hat{\vartheta}$ is the method-of-moments estimator of ϑ if it is a solution of the system of equations $M_l(\vartheta) = \hat{M}_l$, $l = 1..k$.

7 Hypothesis testing

We set up a hypothesis, and examine whether it is true. We accept or reject it. It can be parametric or non-parametric; we can examine the equality of expected values, of standard deviations, their value, the independence of complete event systems. With a goodness-of-fit test we can determine whether the random variables have a given distribution function, and with a homogeneity test whether two samples have the same distribution.

H_0 : null hypothesis, $\vartheta \in \Theta_0$; H_1 : alternative hypothesis, $\vartheta \in \Theta_1$; $\Theta = \Theta_0 \cup \Theta_1$.

One- and two-sided tests: For a two-sided alternative hypothesis we assume non-equality, for a one-sided one some relation. For a two-sided test we examine the absolute value of the test value, whether it is within the acceptance

region; then, for example, for the u-test the given error percentage has to be halved for the computation, since the Φ function is symmetric about the y axis.

- *Statistical test:* $\Xi = \Xi_e \cup \Xi_k$ (disjoint sets) acceptance and critical region. This decomposition is the statistical test. If the observation is an element of the critical region, then we reject the null hypothesis; if it is not an element, then we accept it. $T(\underline{x}) = \begin{cases} 1 & x \in \Xi_k \\ 0 & \text{otherwise} \end{cases}$
- *Type I error:* H_0 is true, but we reject it. Its probability: $P_{\vartheta}(\underline{\xi} \in \Xi_k), \vartheta \in \Theta_0$.
- *Type II error:* H_0 is false, but we accept it. Its probability: $P_{\vartheta}(\underline{\xi} \notin \Xi_k), \vartheta \in \Theta_1$.
The goal is to make these errors as small as possible. The two probabilities can be improved at each other's expense, if the number of observations is fixed.
- *Size of the test:* α is the size of the test if $P_{\vartheta}(\underline{\xi} \in \Xi_k) \leq \alpha, \vartheta \in \Theta_0$.
 α is the exact size of the test if $\sup_{\vartheta \in \Theta_0} P_{\vartheta}(\underline{\xi} \in \Xi_k) = \alpha$.

8 Classical statistical tests

- *u-test:* We assume that the sample is normally distributed ($\xi_i \sim N(m, \sigma^2)$), $i = 1..n$, and that the standard deviation is known.
 - *One-sample:* The null hypothesis is whether the expected value equals a concrete value (m_0); in other words we examine whether the sample mean does not differ significantly from m_0 . So $H_0 : m = m_0$, and in the two-sided case $H_1 : m \neq m_0$, in the one-sided case for example $H_1 : m \geq m_0$ or $H_1 : m < m_0$.
The value of the u-test: $u = \sqrt{n} \frac{\bar{\xi} - m_0}{\sigma}$. If the null hypothesis is true, then this is approximately standard normally distributed.
We examine the hypothesis with error probability ε ; for this we need the value $\Phi(u_{1-\varepsilon}) = 1 - \varepsilon$.
In the two-sided case we reject H_0 if $|u| > u_{1-\frac{\varepsilon}{2}}$, and accept it if $|u| \leq u_{1-\frac{\varepsilon}{2}}$.
In the one-sided case we examine the cases $u > u_{1-\varepsilon}$ (right) and $u < u_{1-\varepsilon}$ (left); in these cases we reject H_0 .
 - *Two-sample:* Here the conditions are the following: $\xi_i \sim N(m_1, \sigma_1^2)$, $i = 1..n$ and $\eta_j \sim N(m_2, \sigma_2^2)$, $j = 1..m$. The standard deviations are likewise known. $H_0 : m_1 = m_2$, and $u = \frac{\bar{\xi} - \bar{\eta}}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}}$. $H_1 : m_1 > m_2$ is the upper (right) sided, and $H_1 : m_1 < m_2$ the lower (left) alternative hypothesis.
- *t-test:* In this test the standard deviation is not known, but we assume a normal distribution just as for the u-test. $\xi_i \sim N(m, \sigma^2)$, $i = 1..n$.
 - *One-sample:* $H_0 : m = m_0$. The alternative hypothesis similarly to the u-test. $t = \sqrt{n} \frac{\bar{\xi} - m_0}{\sqrt{\sigma_*^2}}$, where σ_*^2 is the corrected empirical variance, which we can compute from the sample. (Remark: instead of n we divide by $n - 1$ in the formula.) This value is t -distributed under H_0 , with $n - 1$ degrees of freedom. This test is also called the Student's test.
 - *Two-sample:* $\xi_i \sim N(m_1, \sigma_1^2)$, $i = 1..n$ and $\eta_j \sim N(m_2, \sigma_2^2)$, $j = 1..m$. In this case too the standard deviation is not known, but we assume that the standard deviations of the two samples are equal. Then $t_{n+m-2} = \sqrt{\frac{nm(n+m-2)}{n+m}} \frac{\bar{\xi} - \bar{\eta}}{\sqrt{\sum (\xi_i - \bar{\xi})^2 + \sum (\eta_j - \bar{\eta})^2}}$. $n + m - 2$ is the degrees of freedom of the test.
- *F-test:* Usable for two samples. This test is suitable for examining the equality of standard deviations, so here $H_0 : \sigma_1 = \sigma_2$. If the variances of the two samples are equal, then their quotient tends to 1. $f_{n-1, m-1} = \max(\frac{\sigma_1^2}{\sigma_2^2}, \frac{\sigma_2^2}{\sigma_1^2})$. Of the two degrees of freedom the first belongs to the sample in the numerator of f (-1), the second to that in the denominator.
- *Welch test:* Also called the d-test. Similar to the two-sample t-test, but here the equality of the standard deviations does not have to be assumed. Its degrees of freedom can be computed with a complicated formula.

- *Sequential tests:* $V_n = \frac{\prod f_1(x_i)}{\prod f_0(x_i)} = \frac{L_1(\underline{x})}{L_0(\underline{x})}$. f_0 is the density function under the null hypothesis, f_1 that under the alternative hypothesis. Given a value A and a value B , $A < B$. If $V_n \geq B$, then we reject H_0 ; if $V_n \leq A$, then we accept it; and if $A < V_n < B$, then we take a new sample element.
By *Stein's theorem* N is finite with probability 1. $N = \min\{n : V_n \leq A \vee V_n \geq B\}$.
- *Quality control:* We look at n_1 elements, with thresholds $c_1 < c_2$ and c_3 . If $X_1 \leq c_1$, then we accept H_0 ; if $X_1 \geq c_2$, then we reject it. If $c_1 < X_1 < c_2$, then we look at n_2 elements, and if $X_1 + X_2 \leq c_3$, then we likewise accept H_0 . The expected sample size measures its efficiency.
- χ^2 -test: $H_0 : A_1, \dots, A_n$ a complete event system. $P(A_i) = p_i, i = 1..n$, ν_i the frequency. If the null hypothesis holds, then $\frac{\nu_i}{n} \sim p_i$.
 $\chi^2 = \sum \frac{(\nu_i - np_i)^2}{np_i}$. This is χ^2 -distributed, with $r - 1$ degrees of freedom. r is the number of groups added together. (Remark: if the frequency in some group were too small, then we merge those.)
The χ^2 -test can be used for goodness-of-fit, homogeneity and independence testing. (Remark: there is a different formula for each.)
- *Other tests:*
 - *Kolmogorov–Smirnov test:* whether 2 empirical distribution functions are equal (homogeneity test), or, for 1 sample, whether it equals some distribution function. $D_{m,n} = \max_x |F_n(x) - G_m(x)|$. X_i with distribution function F , Y_j with G . $H_0 : F \equiv G$.
 - *Sign test:* How many times it holds that something is positive.
 - *Wilcoxon test:* (rank statistic), to test $P(X > Y) = \frac{1}{2}$ we count for how many pairs it holds that $X_i > Y_j$.